

June 18, 2026

To the Editorial Board
Physical Review Research

Dear Editors,

Please consider the enclosed manuscript, “A two-feedback Vicsek–Couzin flock: dense-phase noise rectification as the unifying action of density-gated motility–noise coupling,” for publication in *Physical Review Research* as an original research article.

The standard Vicsek model fixes the self-propulsion speed and the angular-noise distribution as global constants. Two relaxations have been studied separately: a density-dependent speed, which drives motility-induced phase separation, and heavy-tailed Lévy reorientations, which reshape the order–disorder transition. We ask what happens when both are switched on together, gated by a single density-dependent sigmoid in the local neighbour count, and we isolate each contribution with a four-way ablation (baseline / noise-shape-only / motility-only / full) run on one code path.

We find that heavy-tailed noise on its own grows no size-dependent susceptibility; only the motility channel builds a phase-separated dense phase. The noise-shape channel then acts inside that dense phase through a single route, rectifying its heavy-tailed Cauchy kicks into Gaussian ones. What at first looks like a super-additive cooperation between the two channels we show, with a matched-noise control, to be one effect: equalising the effective dense decorrelation collapses the susceptibility excess (Δ_9 from +0.45 to +0.02) and reverses the static $g(r)$ and dynamic $C(\tau)$ gaps, so the two channels are additive and the noise-shape feedback simply lowers the dense effective noise. The robust results are a clean two-regime split that reproduces across three densities, a density-gated Lévy-to-Gaussian conversion of dense-phase turning, and a decoupled-sigmoid dissection that pins the motility threshold as the sole control variable and shows the effect to be a property of the metric kernel.

We believe the work suits *Physical Review Research* well. It is a careful, well-controlled study of an active-matter model (Vicsek and Couzin flocking, motility-induced phase separation, giant number fluctuations, finite-size scaling of the order–disorder transition, and density-stratified correlation functions) whose central result is a clarifying, rigorously controlled one. The journal’s broad, open-access remit and its receptiveness to solid unifying and negative results make it a natural home, and the work should interest the active-matter, soft-matter and non-equilibrium statistical-physics communities the journal serves.

The analysis is deliberately conservative. Claims rest on ten-seed finite-size scans with seed-bootstrap confidence intervals and non-parametric (Wilcoxon signed-rank, Mann–Whitney) tests rather than Gaussian z -scores; null results (a passive noise-shape threshold, negligible hysteresis, no extended patch lifetime, a topological-kernel breakdown, and a falsified mean-density prediction) are reported rather than buried; the finite-size slopes are presented as crossover rates, not asymptotic exponents, dropping as the lever is extended to nine sizes (L up to 256); and the apparent synergy is shown, by the matched-noise control, to be additive at matched noise rather than an emergent cooperation. All code and data are openly available at <https://github.com/yayayou47/two-feedback-vicsek> for bit-for-bit reproduction of every figure.

This manuscript is original, has not been published previously, and is not under consideration at any other journal. It has a single author, who declares no competing interests. One supplementary-material PDF accompanies the submission. As potential referees, we would suggest specialists in active-matter and flocking models, in finite-size scaling of non-equilibrium phase transitions, and in motility-induced phase separation.

Thank you for considering our submission.

Sincerely,

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